

ORGANIC ELECTRONICS STARTS A NEW ERA

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Organic microdisplays for ultra-portable digital handhelds

■ PRABHAT K. MISHRA,
SAHAJ SAXENA

Organic electronics, or plastic electronics, is a branch of electronics that deals with conductive polymers which are carbon based. Inorganic electronics, on the other hand, relies on inorganic conductors like copper or silicon.

For the past four decades, inorganic semiconductors (Si, Ga, As) and metals (Cu, Al) have been the backbone of the electronics industry. But, if we review the growth of the electronics industry, it is clear that innovative organic materials play a vital role in the per-

formance increase of semiconductors, storage and display at lower costs that we see today. Organic electronic products are lighter, more flexible and less expensive than their inorganic counterparts. These are also biodegradable (being made from carbon). This opens the door to many exciting and advanced new applications that would be impossible using copper or silicon.

However, conductive polymers have a high resistance and therefore are not good conductors of electricity. In many cases, these also have shorter lifetimes and are much more dependent on stable environment conditions

than inorganic electronics.

Devices using organic compounds

Some of the major devices that are manufactured using organic compounds are organic thin-film transistors (OTFTs), organic light-emitting diodes (OLEDs) and organic solar cells.

TFTs are transistors created using thin films, usually of silicon deposited on glass. The deposited silicon must be crystallised using laser pulses at high temperatures.

The active layers of OTFTs can be thermally evaporated and deposited on any organic substrate (a flexible piece of plastic) at much lower temperatures. OTFTs are used to design complex architectures like arithmetic logic units and non-linear oscillators, active-matrix displays, chemical sensors and flexible microelectronics.

The structure of an organic semiconducting p-type thin-film transistor with top contacts is shown in the figure.

An OLED is a thin-film LED in which the emissive layer is an organic compound. Conventionally, an OLED has a bottom emitting structure, which includes an opaque metal or metal alloy cathode, and a transparent anode on a transparent substrate enabling light to be emitted from the bottom of the structure usually from the glass side.

Organic solar cells could cut the cost of solar power by replacing the expensive crystalline silicon used in most solar cells with inexpensive organic polymers. What's more, the polymers